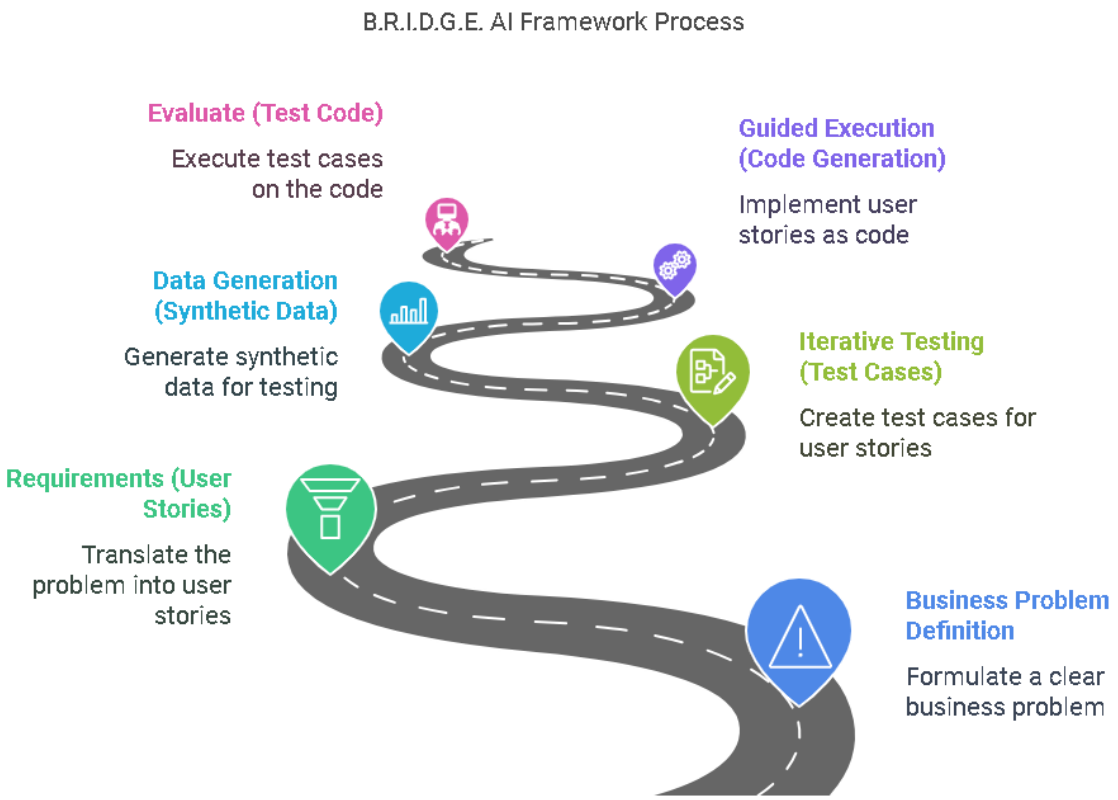
BRIDGE AI Framework

(Business Problem, Requirements, Iterative Testing, Data Generation, Guided Execution & Evaluation using AI)

BRIDGE. AI Framework Overview

Purpose: This framework provides a systematic process for developing applications or projects using AI Assistant. Starting from a business problem, AI Assistant iteratively generates each component (user stories, test cases, synthetic data, code, etc.), verifies each step against defined metrics, and refines outputs based on validation results. AI Assistant acts as both a creator and a validator, enhancing each step's quality and alignment with the project's objectives.



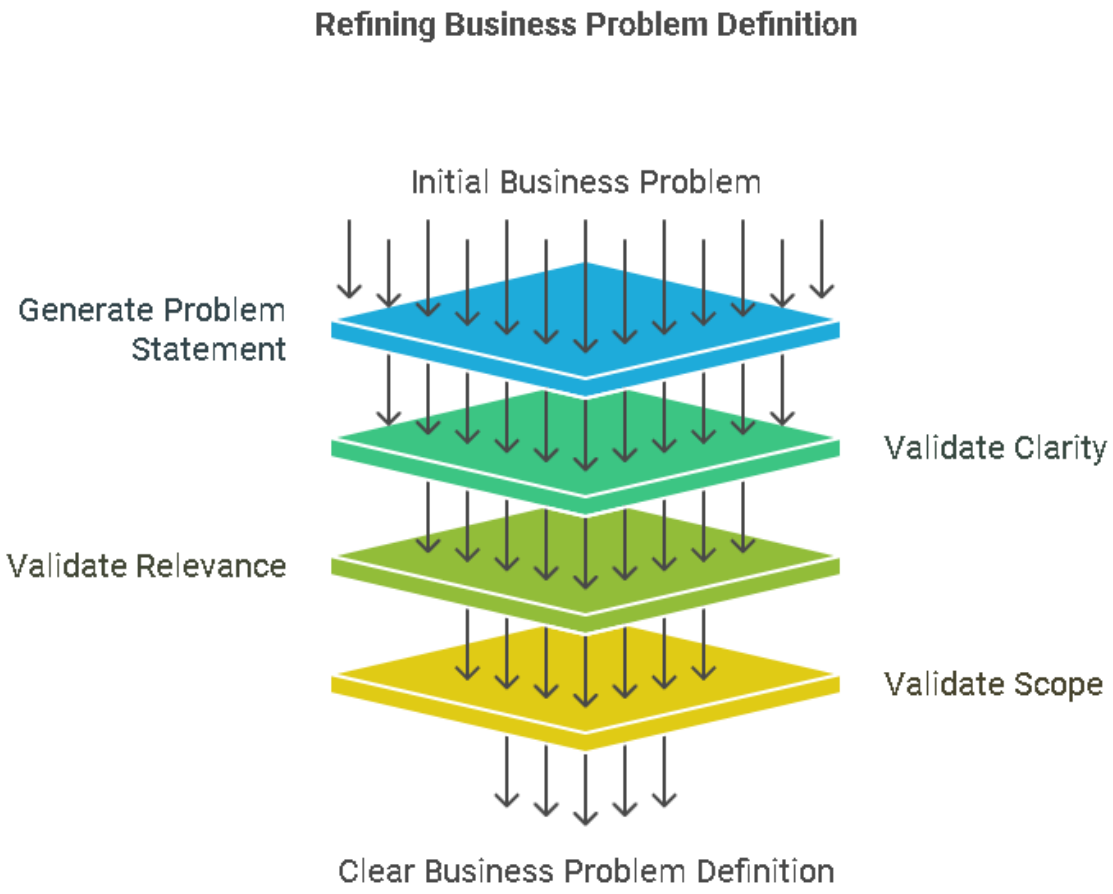
Step 1: Business Problem Definition

Objective: Formulate a clear business problem or scenario that defines the project's purpose, target outcomes, and necessary functionalities.

- **Metrics for Success**:
  - **Problem Clarity**: The problem statement is specific, clear, and well-defined.
  - **Objective Relevance**: Confirm alignment with measurable business objectives.
  - **Scope Definition**: Check that the scope is manageable, with key features identified.

AI Assistant Role:

1. **Generate**: User provides the initial business problem, which AI Assistant refines for clarity, if necessary.
2. **Validate**: AI Assistant checks the statement against metrics for clarity, relevance, and scope, suggesting improvements if needed.



Master Prompt:

The specific task or goal is to formulate a clear business problem or scenario that defines the project's purpose, target outcomes, and necessary functionalities. **<High Level problem statement goes here**>. AI Assistant should adapt following roles:

- Generate: AI Assistant generates the Business problem definition based on information provided.
- Validate: AI Assistant checks the statement against metrics for clarity, relevance, and scope, suggesting improvements if needed

It should use following success metrics :

- Problem Clarity: The problem statement is specific, clear, and well-defined.
- Objective Relevance: Confirm alignment with measurable business objectives.
- Scope Definition: Check that the scope is manageable, with key features identified

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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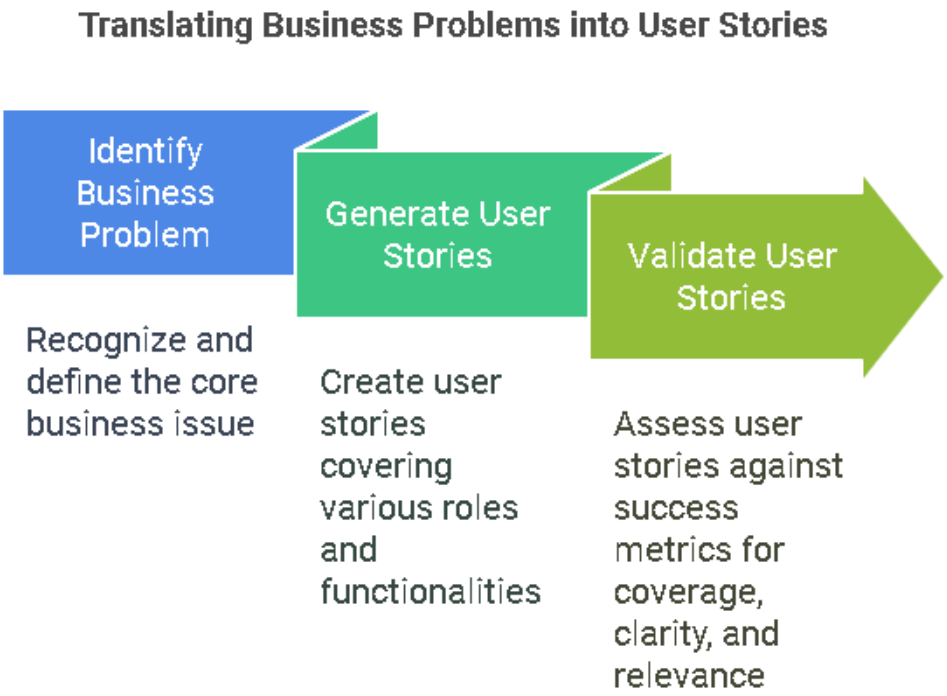
Step 2: Requirements (User Stories)

Objective: Translate the business problem into specific user stories that outline the essential functionalities from the end user's perspective along with acceptance criteria .

- **Metrics for Success**:
  - **Coverage**: Ensure each critical functionality from the business problem is captured in user stories.
  - **Clarity and Specificity**: User stories are clear, specific, and free from ambiguity.
  - **Relevance**: Each user story directly contributes to the objectives of the business problem.

AI Assistant Role:

1. **Generate**: AI Assistant generates user stories based on the business problem, covering various user roles and functionalities.
2. **Validate**: AI Assistant evaluates user stories against metrics, checking coverage, clarity, and relevance. If any metric isn't met, AI Assistant revises and improves the user stories accordingly.



Master Prompt:

The specific task or goal is to translate the business problem into specific user stories that outline the essential functionalities from the end user's perspective along with acceptance criteria.

The business problem is -<Business Problem Definition as context goes here>

AI Assistant should adapt following roles:

- Generate: AI Assistant generates the user stories based on business problem definition covering various user roles and functionalities.
- Validate: AI Assistant checks the statement against success metrics, suggesting improvements if needed.

It should use following success metrics :

- Coverage: Ensure each critical functionality from the business problem is captured in user stories.
- Clarity and Specificity: User stories are clear, specific, and free from ambiguity.
- Relevance: Each user story directly contributes to the objectives of the business problem.

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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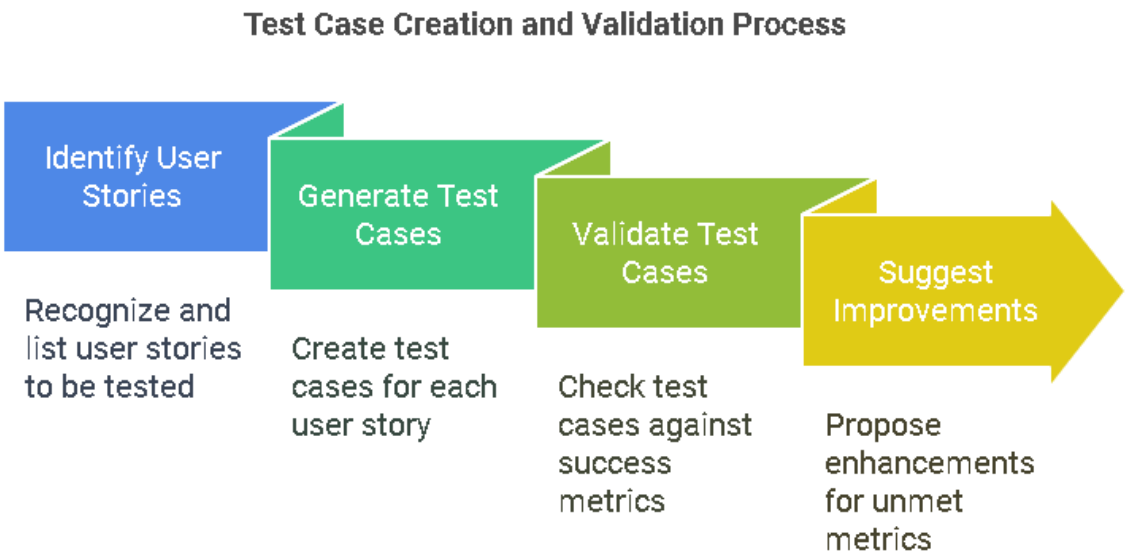
Step 3: Iterative Testing (Test Cases)

Objective: Create test cases that validate each user story, ensuring functionalities meet requirements, including both normal and edge cases.

- **Metrics for Success**:
  - **Completeness**: Each user story has at least one test case validating its primary functionality.
  - **Validity**: Test cases realistically simulate user interactions and potential errors.
  - **Coverage**: Edge cases and error handling scenarios are included.

AI Assistant Role:

1. **Generate**: AI Assistant generates test cases for each user story, covering success and failure scenarios.
2. **Validate**: AI Assistant checks the test cases against the metrics, ensuring completeness, validity, and coverage. If any metric is unmet, AI Assistant suggests improvements or additional test cases.



3.

Master Prompt:

The specific task or goal is to create test cases that validate each user story, ensuring functionalities meet requirements, including both normal and edge cases.

The user stories are -<User story goes here>

AI Assistant should adapt following roles:

- Generate: AI Assistant generates the test cases based on user story provided covering success and failure scenarios.
- Validate: AI Assistant checks the test cases against success metrics, If any metric is unmet, AI Assistant suggests improvements or additional test cases

It should use following success metrics :

- Completeness: Each user story has at least one test case validating its primary functionality.
- Validity: Test cases realistically simulate user interactions and potential errors.
- Coverage: Edge cases and error handling scenarios are included.

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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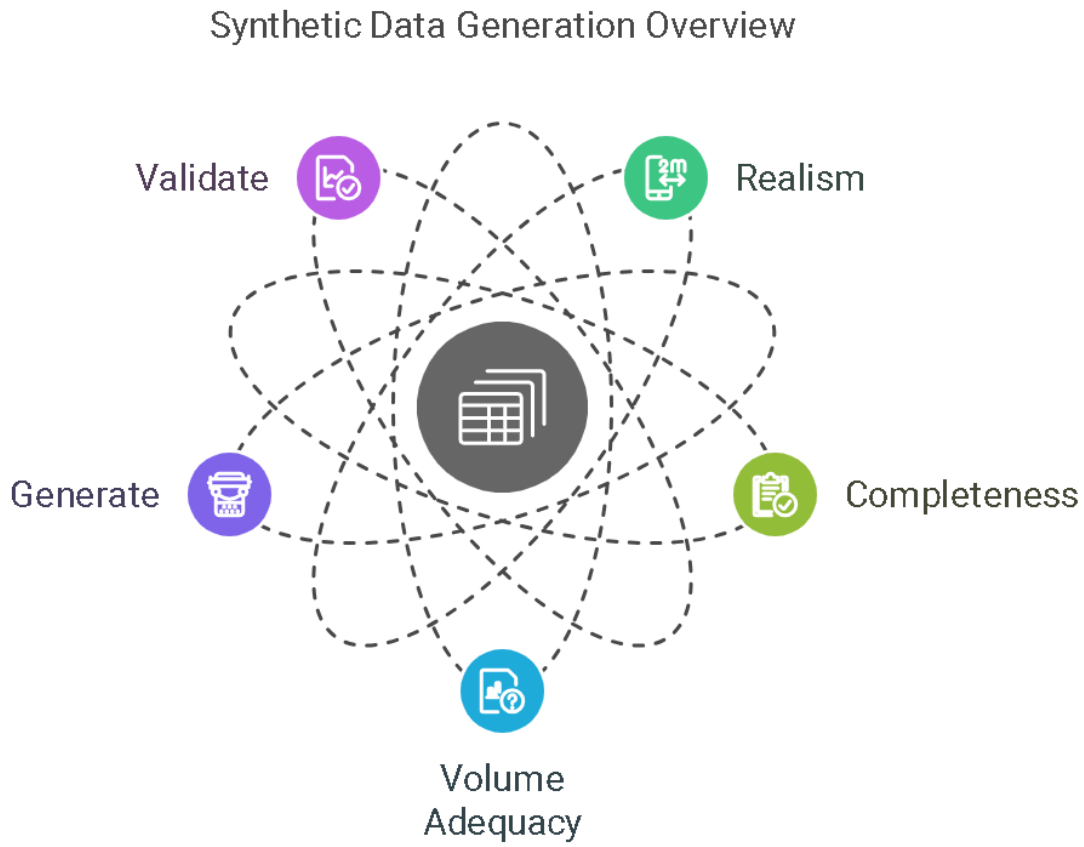
Step 4: Data Generation (Synthetic Data)

Objective: Create synthetic data that closely resembles real-world data for testing purposes, covering all necessary fields and scenarios.

- **Metrics for Success**:
  - **Realism**: Synthetic data mimics real-world data in structure, format, and diversity.
  - **Completeness**: Data includes all fields required by user stories and test cases.
  - **Volume Adequacy**: Data volume is sufficient to test system performance under typical and high-load scenarios.

AI Assistant Role:

1. **Generate**: AI Assistant generates synthetic data, simulating real-world inputs and ensuring variation.
2. **Validate**: AI Assistant evaluates the synthetic data against the metrics, checking realism, completeness, and volume adequacy. If needed, AI Assistant provides suggestions to improve data realism or add missing fields.



Master Prompt:

The specific task or goal is to create synthetic data that closely resembles real-world data for testing purposes, covering all necessary fields and scenarios.

The test cases are -<test cases goes here>

AI Assistant should adapt following roles:

- Generate: AI Assistant generates synthetic data, simulating real-world inputs and ensuring variation.
- Validate: AI Assistant evaluates the synthetic data against the metrics, checking realism, completeness, and volume adequacy. If needed, AI Assistant provides suggestions to improve data realism or add missing fields.

It should use following success metrics :

- Realism: Synthetic data mimics real-world data in structure, format, and diversity.
- Completeness: Data includes all fields required by user stories and test cases.
- Volume Adequacy: Data volume is sufficient to test system performance under typical and high-load scenarios.

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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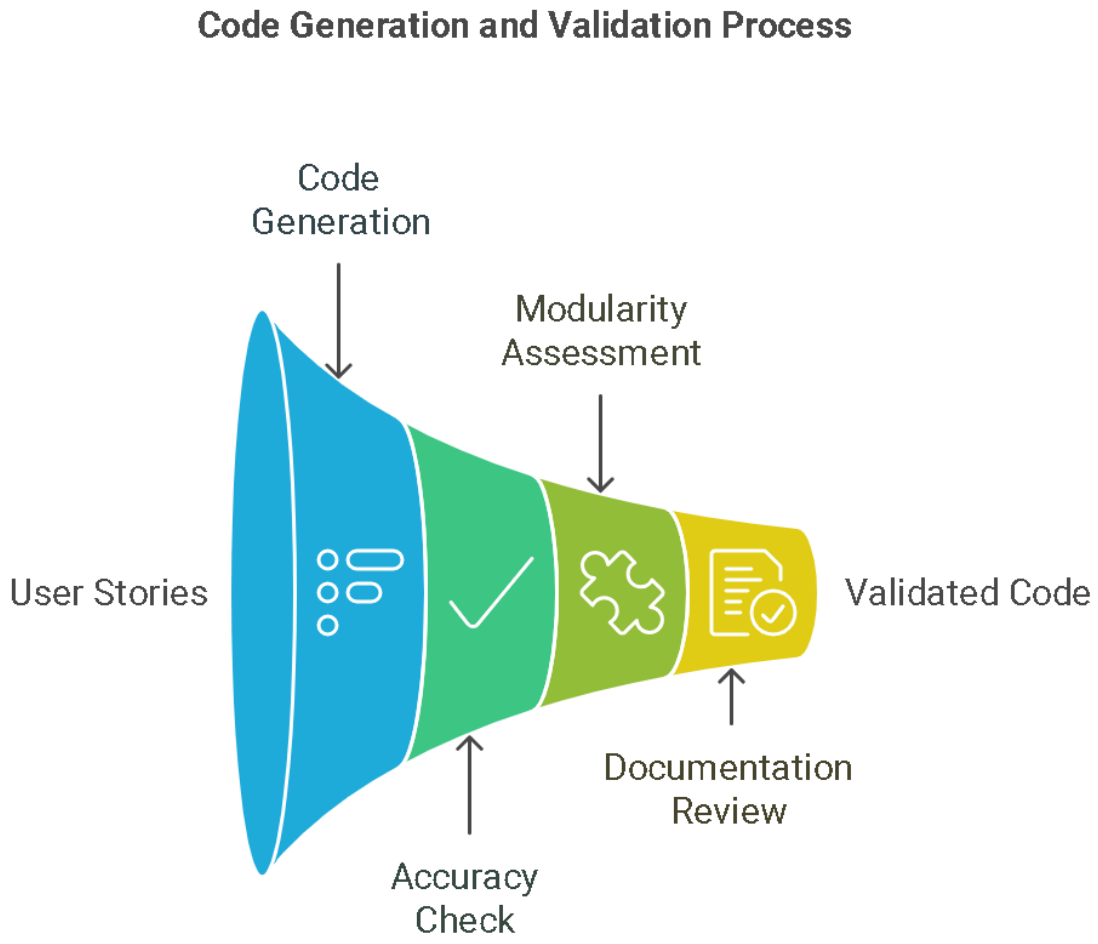
Step 5: Guided Execution (Code Generation and Validation)

Objective: Implement each user story as code, ensuring each functionality aligns with business requirements and passes the associated test cases.

- **Metrics for Success**:
  - **Accuracy**: Code meets each user story's functional requirements and passes test cases.
  - **Modularity**: Code is organized into independent functions or classes, supporting reusability.
  - **Documentation**: Code includes comments explaining logic, purpose, and usage for readability.

AI Assistant Role:

1. **Generate**: AI Assistant generates code for each user story one step at a time, structuring it for modularity and readability.
2. **Validate**: AI Assistant validates the code against metrics, running test cases (or providing pseudocode for them) and verifying modularity and documentation. If any metric is unmet, AI Assistant refines the code and documentation.



Master Prompt:

The specific task or goal is to Implement each user story as code, ensuring each functionality aligns with business requirements and passes the associated test cases.

The test cases are -<test cases goes here>

AI Assistant should adapt following roles:

- Generate: AI Assistant generates code for each user story one step at a time, structuring it for modularity and readability.
- Validate: AI Assistant validates the code against metrics, running test cases (or providing pseudocode for them) and verifying modularity and documentation. If any metric is unmet, AI Assistant refines the code and documentation

It should use following success metrics :

- Accuracy: Code meets each user story's functional requirements and passes test cases.
- Modularity: Code is organized into independent functions or classes, supporting reusability.
- Documentation: Code includes comments explaining logic, purpose, and usage for readability

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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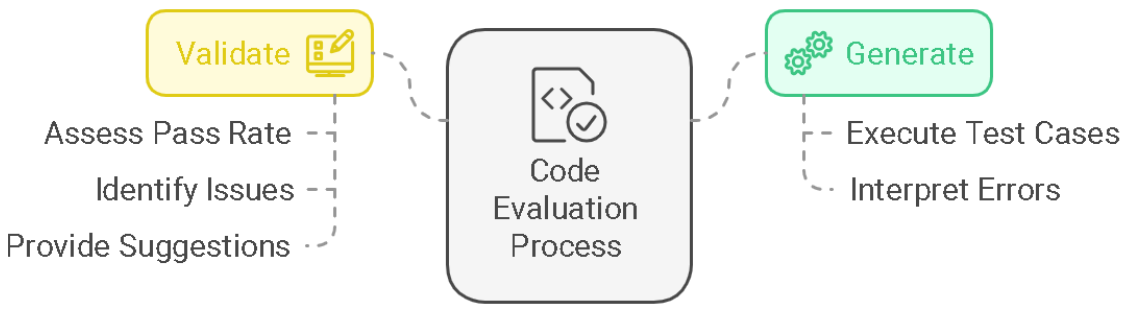
Step 6: Evaluate (Test Code with Test Cases)

Objective: Execute test cases on the generated code to verify each functionality's performance, identifying and resolving any issues.

- **Metrics for Success**:
  - **Pass Rate**: A high percentage of test cases pass, demonstrating functionality correctness.
  - **Error Identification**: Any failing test cases are documented for debugging.
  - **Resolution Efficiency**: Track the time taken to fix errors, indicating code quality and ease of maintenance.

AI Assistant Role:

1. **Generate**: AI Assistant assists in executing test cases and helps interpret any errors that arise.
2. **Validate**: AI Assistant assesses the pass rate of test cases and identifies issues for debugging, providing suggestions for code refinement or bug resolution.



Master Prompt:

The specific task or goal is to Execute test cases on the generated code to verify each functionality's performance, identifying and resolving any issues s.

Please use test cases and generated code from previous steps

AI Assistant should adapt following roles:

- Generate: AI Assistant assists in executing test cases and helps interpret any errors that arise.
- Validate: AI Assistant assesses the pass rate of test cases and identifies issues for debugging, providing suggestions for code refinement or bug resolution.

It should use following success metrics :

- Pass Rate: A high percentage of test cases pass, demonstrating functionality correctness.
- Error Identification: Any failing test cases are documented for debugging.
- Resolution Efficiency: Track the time taken to fix errors, indicating code quality and ease of maintenance

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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Step 7: AI-Driven Dashboard (Visualization)

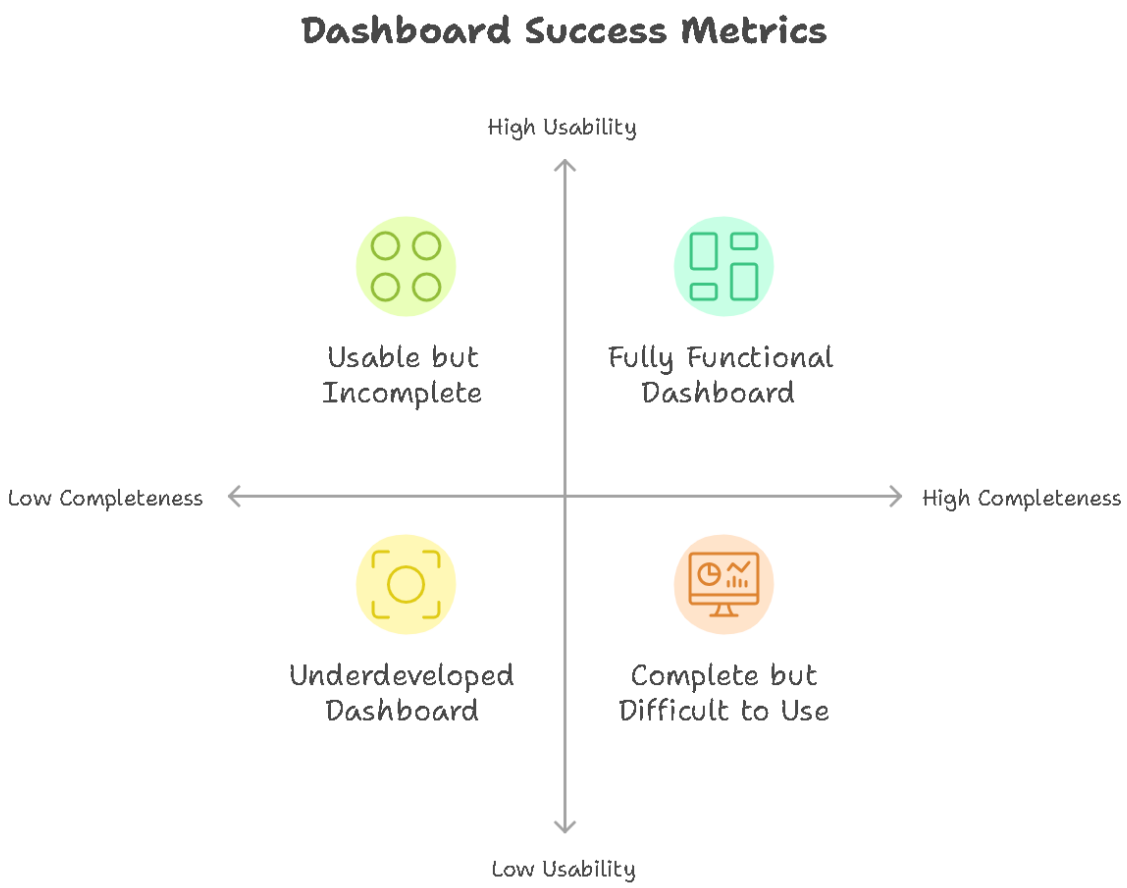
Objective: Develop a simple web application or dashboard (e.g., with Flask) to present the project results, providing a user-friendly interface and real-time metrics.

- **Metrics for Success**:
  - **Completeness**: The interface includes all functionalities, accessible for user interaction.
  - **Usability**: The interface is intuitive, allowing seamless interaction with features.
  - **Metric Accuracy**: Key metrics (e.g., total books, borrowed books, availability) are displayed accurately, providing data-driven insights.

AI Assistant Role:

1. **Generate**: AI Assistant generates code for the dashboard or web interface, setting up pages, routes, and visualizations.

2. **Validate**: AI Assistant verifies that the interface is user-friendly, functional, and displays data accurately. It can simulate user interactions or suggest testing methods for further validation.



Master Prompt :

The specific task or goal is to Develop a simple web application or dashboard (e.g., with Flask) to present the project results, providing a user-friendly interface and real-time metrics.

Please use generated code from previous steps

AI Assistant should adapt following roles:

- Generate: AI Assistant generates code for the dashboard or web interface, setting up pages, routes, and visualizations.
- Validate: AI Assistant verifies that the interface is user-friendly, functional, and displays data accurately. It can simulate user interactions or suggest testing methods for further validation.

It should use following success metrics :

- Completeness: The interface includes all functionalities, accessible for user interaction.
- Usability: The interface is intuitive, allowing seamless interaction with features.
- Metric Accuracy: Key metrics (e.g., total books, borrowed books, availability) are displayed accurately, providing data-driven insights.

Also remember that first only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality

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End-to-End Metrics Summary with AI Assistant Validation

After completing each step and validating individual metrics, AI Assistant can help assess the entire project against the following final criteria:

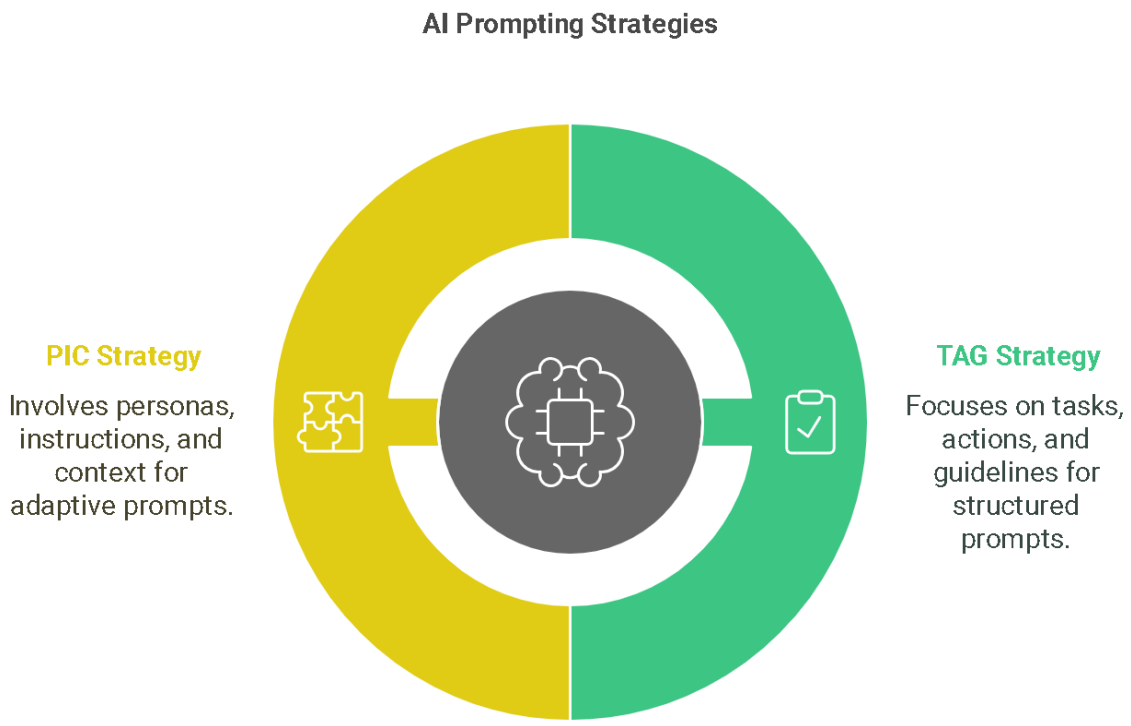
1. **Overall Completeness**: Ensure that each user story is implemented and accessible from the interface.
2. **Feature Accuracy**: Confirm that each feature functions as expected, based on test cases and metrics.
3. **Data Validity**: Verify that synthetic data accurately represents real-world data, providing a reliable testing foundation.
4. **Usability and Relevance**: Ensure the final interface is user-friendly and aligns with the business problem.

Workflow Summary with AI Assistant

1. **User-Defined Start**: The user provides the initial business problem or scenario.
2. **Step-by-Step Generation and Validation**: AI Assistant generates each step (user stories, test cases, synthetic data, code, etc.) and validates outputs using defined metrics, refining each component as needed.
3. **Iterative Refinement**: AI Assistant assists in refining each output to meet success criteria before moving to the next step.
4. **Final Integration**: AI Assistant helps integrate all components into a cohesive application or dashboard that fulfils the business problem requirements.

The **B.R.I.D.G.E. AI** framework leverages AI Assistant's capabilities to build, validate, and refine each component of the project. This structured, metric-driven approach ensures each step aligns with the business problem and results in a reliable, user-friendly application.

Prompting Strategies



Strategy 1 - Task-Actions-Guidelines (TAG) Prompt Strategy :

1.1 - Task: The specific task or goal is to <provide the objective for the step here >

1.2 - Actions: It should cover these actions or required functionality < Actions goes here>

1.3 - Guidelines: Provide the success metrics for the step here <Success Metrics>

Strategy 2 – Persona-Instructions-Context (PIC) Prompt Strategy:

2.1- Persona: For the Bridge AI framework . we have fixed set of roles. Hence AI Assistant should adapt following roles:

- Generate: <provide generator expectations here>.
- Validate: AI Assistant checks the statement against success metrics, suggesting improvements if needed:

2.2 - Instructions: For the BRIDGE AI framework always include these instructions - First only provide the output for Generate step and wait for user feedback before proceeding to the validate the input using validate persona and stick to the information provided for functionality don't add any extra functionality.

2.3 - Context: Provide the output from previous steps here as context if required. For e.g. output from Problem definition goes as Context for the User stories.